

Counterparty Credit Risk Engine for the Capitolis Book

Data, Models, Trade-offs, and Results

Berkeley MFE Industry Project – Capitolis

Valuation date: 2026-08-28 | September 30, 2026

Agenda

Part I: The Book and the Mandate

Part II: Risk Factors and Data

Part III: Risk-Factor Models and Calibration

Part IV: Simulation Methodology

Part V: Engine, Equations, and Headline Results

Part VI: Sensitivities, Stress, and Validation

Part VII: Credit and Capital ($\times$ VA)

Part VIII: Decision Register and Limitations

Part I

What Capitolis Gave Us

The book: 16 trades, three netting sets

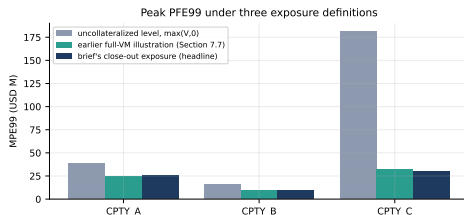
Trade	Type	Cpty	Underlying	Ends	Notional	NPV $t=0$ (USD)
EQTRS_0001-3	Eq. TRS	A	3 USD names each	Oct'26	50-99M	-5.55M / -1.08M / +9.92M
EQTRS_0004	Eq. TRS	B	16 USD names	Nov'26	49.2M	-0.87M
EQTRS_0005/6	Eq. TRS	B	3 JPY names each	Oct'26	12-31M	+0.15M / +1.63M
EQTRS_0007/8	Eq. TRS	C	7 / 3 USD names	Oct'26 / Jul'27	95M / 30M	-0.37M / +2.22M
BF_0001/2	Bond fwd	A	UST 3.88%'28 / 4.63%'35	Oct/Nov'26	100M / 20M	+0.51M / -0.36M
BF_0003	Bond fwd	C	UST 2.88% 5/15/2049	Dec'26	500M	+121.26M
BF_0004	Bond fwd	C	UST 1.13%'28	Oct'26	50M	+1.97M
BTRS_0001-4	Bond TRS	A/B/B/C	UST 4.12/2.88/4.63/4.12%	'27-'28	5-40M	+0.12M / +0.21M / +0.03M / +0.04M

Net MtM by netting set: CPTY_A +\$3.57M CPTY_B +\$1.15M CPTY_C +\$125.12M **Book +\$129.84M**

The exposure we must measure (Capitolis kickoff definition)

$$\max(V(t+10bd) - V(t-1bd), 0)$$

- **Variation margin:** netted value one business day before t , two-way, no threshold
- **Margin period of risk (MPoR):** close-out 10 business days after t
- **Netting:** within a counterparty only – book exposure = sum over netting sets; *never* netted across counterparties
- **Trades settling inside the window** are excluded from both legs (a scheduled settlement is cash, not a market move)



Part II

Risks, Variables, and Data

Risk factors: what can move the book

Risk factor	Trades affected	Model	Calibrated from
37 equity prices	All 8 equity TRS	Correlated geometric Brownian motion	3y realised vol, Yahoo Finance dividend yield
USDJPY	Equity TRS on JPY names (2)	GBM, quanto-adjusted drift	3y realised FX vol
USD short rate (whole curve)	Bond forwards, bond TRS, funding legs, discounting	Hull-White one-factor (equivalent to a 1-factor LGM)	Real swaption cube (a) + realised long-end Treasury vol (σ)
JPY short rate	Drift of JPY-listed names and USDJPY	A second, independent Hull-White one-factor	Real JPY OIS curve, JPY rate history
Counterparty credit	CVA/DVA/FVA/KVA only	Deterministic hazard-rate survival curve	ICE BofA rating-index OAS proxy; no CDS available

39 correlated stochastic factors, one joint simulation: 37 equities, USDJPY, USD short rate, JPY short rate – correlated by a single Cholesky factor of the real historical matrix at every time step.

- **Every number in this project traces to a disclosed, real source** – no synthetic placeholders for calibration inputs
- **Benefit:** results are reproducible, auditable, and defensible in front of a risk committee; every assumption is a named simplification, not a silent one
- Two tiers: **public** (Yahoo Finance, FRED, Databento) and **licensed** (Bloomberg, used and cross-checked but never redistributed as raw series)

Data source 1: USD discount curve (Databento SOFR futures)

- **Source:** real Databento CME SOFR futures settlement prices (SR3), bootstrapped ourselves
- **Used for:** every USD discount factor; the base for the Hull-White short-rate model
- **Extension:** Bloomberg long-end quotes spliced in beyond futures coverage (out to 50y), since SOFR futures alone only cover the front few years

Benefit of real data: a genuine market-implied curve, not an assumed flat extrapolation. In the correction-by-correction attribution (Part VI), splicing the real Bloomberg long end onto the previously-flat curve moved the portfolio MPE99 from \$52.7M to \$52.1M (−1.1%) – a modest, disclosed effect, not a large one, but a genuinely measured one rather than an assumption

Data source 2: Treasury yields (FRED) and swaption vols (Bloomberg)

Source	Used for	Benefit of using it
FRED Treasury CMT yields (3mo–30y, public, daily since 1981)	Realised long-end yield volatility, used to fit Hull-White σ (0.96%); Kupiec backtest of the rate model	Directly measures the volatility of the yields this book's bond positions actually depend on, not a proxy
Bloomberg ATM normal swaption vol cube	Mean-reversion a (0.0167, $R^2=0.84$), via the decay of vol with tenor	Textbook calibration route for a ; genuinely implied, forward-looking

Data source 3: equities, FX, JPY curve

Source	Used for	Benefit of using it
Yahoo Finance: 37 equity closes + dividends, 3y history	Equity spots, dividend yields, realised volatility, the 39×39 correlation matrix	Public, free, long enough history (2014–today, extended to 2007 for stress) to estimate realised statistics reliably
Yahoo Finance: USD-JPY spot, 3y history	FX spot, realised FX volatility, FX–equity correlation	Same realised-statistics benefit, consistent methodology across all factors
Real JPY OIS curve (public JPY market data)	JPY Hull-White factor's base curve	Lets JPY rates move independently and go negative, matching the real BOJ history

Data source 4: credit spreads, and the licensed Bloomberg data

Source	Used for	Benefit of using it
ICE BofA bond-index OAS by rating (FRED, public)	Counterparty and issuer credit spread proxy (CVA, DVA, risky-bond pricing)	No CDS available; this is the Basel MAR50.32(3)-sanctioned fallback – real market spreads by credit quality, not an as- sumed flat number
Bloomberg (licensed): long-end curve, USD/JPY swaption vol cubes, FX vol sur- face	USD/JPY curve splice, mean-reversion fit, model-risk cross-checks	The only source for genuine long-dated mar- ket data beyond what's public; used inter- nally and only derived numbers are ever re- ported (never raw series), per the license

Benefit, summarised: every model parameter is either directly measured or transparently proxied from real data, so every number in this deck can be traced back to a named, dated market source.

From data to risk: one chain per risk factor

Data	Risk factor	Parameter	Model	Risk output
SOFR futures + Bloomberg long end	USD curve	$P(0, T), \sigma=0.96\%$	Hull-White 1F	Bond fwd / TRS value, DV01
Bloomberg swaption cube	USD rate mean reversion	$a=0.0167$	Hull-White 1F	Rate exposure shape
Real JPY OIS curve	JPY short rate	$a=0.001, \sigma=0.27\%$	Hull-White 1F (JPY)	JPY-name / USDJPY drift
Yahoo Finance equity prices	37 equities	vol 18%–76%	Correlated GBM	Equity TRS value
Yahoo Finance USDJPY spot	USDJPY	vol $\sim 9\%$	GBM, quanto	JPY legs
ICE BofA rating OAS	Counterparty credit	BBB proxy spread	Hazard-rate survival	CVA, DVA, FVA, KVA, SA-CVA

All 39 shocks are correlated by one Cholesky factor at every time step; exposure is aggregated per netting set, never across counterparties.

Part III

Models Considered, and Why the Final Model

Rate models considered

Model	Advantages	Disadvantages	Data required	Final use
Hull-White 1F (equiv. 1-factor LGM)	Exact step, closed-form $P(t, T)$; 2 parameters; stable calibration	One factor – curve moves close to parallel; can't twist	Curve + swaption row + realised vol	Selected Production
G2++ (2-factor Gaussian, equiv. LGM2F)	Level and slope move separately; fits the realised covariance	5 parameters; b hits its upper bound; 16.1% relative fit error	Curve + full realised yield covariance	Challenger, tested
2-factor + stochastic vol	Vol clustering, fatter tails	Not implemented; more parameters, no closed-form calibration here	Swaption smile / history	rejected not built

The question is empirical, not a matter of preference: does the second factor materially change this book's exposure? Tested head to head in the next 3 slides.

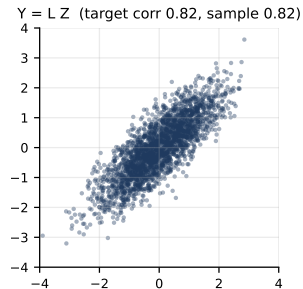
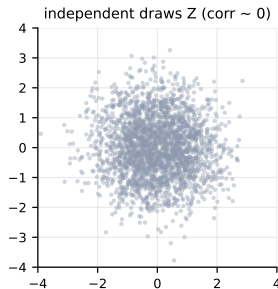
Equity, FX and credit models considered

Model	Advantages	Disadvantages	Final use
Equities: GBM	Exact log-Euler step, closed-form drift, fast	No vol smile or clustering	Selected Production
Equities: stochastic vol (Heston)	Vol clustering, skew	Not implemented; 5+ extra parameters per name, needs option surfaces	rejected not built; stress used instead
FX: GBM with quanto/rate-differential drift	Consistent with both simulated curves	No smile	Selected Production
Credit: deterministic hazard-rate survival (rating proxy)	Transparent, Basel MAR50.32-compliant	No wrong-way risk, no spread volatility	Selected Production
Credit: stochastic intensity	Spread risk, wrong-way risk	No counterparty spread history / CDS available	rejected not available

Why simple models where the book is linear and short-dated: exposure windows are 10 business days on largely linear TRS/forward payoffs; the stress-testing programme (Part VI) tests the effect of a volatility *regime* shift directly, rather than building it into the base model.

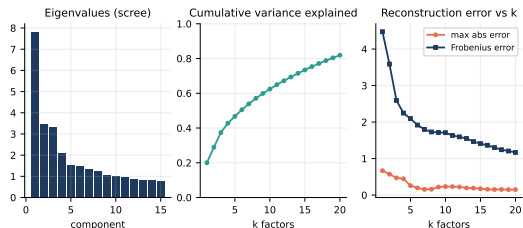
Equity and FX: correlated geometric Brownian motion

- Standard lognormal diffusion, risk-neutral drift ($r - q$), correlated via the real 39×39 matrix
- JPY-quoted names carry a quanto correction; USDJPY drift enforces the martingale condition on the JPY money-market account valued in USD



Cholesky demonstration on the most correlated real pair (corr 0.82): independent draws (left) become correlated (right) after multiplying by L .

Correlation: full Cholesky matrix vs. PCA factor model



		Full	PCA k=5	PCA k=10
Step	time	1.91s	2.08s	2.29s
(5,000 scen.)				
Variance	ex-	100%	47%	62%
plained				
Netting-set	vol	–	B +1.7%	B +1.5%
vs. full				

Selected full matrix: PCA is *slower*, not faster, at $n=39$ (draws k systematic + n idiosyncratic random numbers, no dimensionality win). **Loss of using PCA instead:** no speed gain, and up to 1.7% extra sampling error – a pure loss here. Kept only as an explainability tool: factor 1 (20% of variance) is a broad market mode, factor 2 (8.9%) is Japan vs. US.

USD rates: calibrating mean reversion (a) – two real routes compared

Route	Data	a	R^2	Verdict
Swaption cube (preferred)	Bloomberg ATM normal vol, 1M expiry, 1–15Y tenors	0.0167	0.845	Selected used
SOFR-futures vol decay	~2y Databento history, 8 contracts	0.0458	0.820	cross-check only

Both real-data routes give sensible, positive values with similar fit quality; the swaption route is the textbook standard (implied, not realised) and was preferred. The two disagree because they use different instruments (implied option vol vs. realised futures moves) and different tenor ranges – a real, disclosed source of model uncertainty, quantified directly in the model-risk study ($a \times 3$ moves the headline number by -5%).

USD rates: calibrating volatility (σ) – realised, and which window

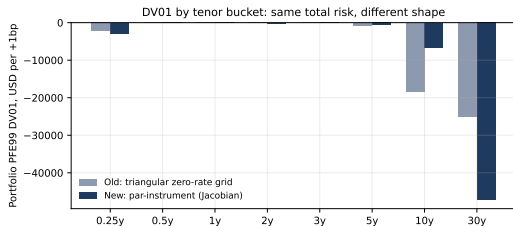
- **Original source:** realised volatility of the overnight SOFR fixing itself, 0.63%
- **Problem:** overnight fixings move in discrete steps on Fed days – understates how far the *long-dated* yields this book depends on actually move (realised 10-day change of the 20y yield: $\sim 15\text{bp}$ std dev, vs. $\sim 11\text{bp}$ implied by the overnight sigma)
- **Selected Changed to:** realised volatility of 2y–30y Treasury yield changes directly, 0.96%

Validated by Kupiec backtest (99% VaR, rate moves, 161 ten-day windows, 3 tenors):

Tenor	Overnight-SOFR σ : up/down exc.	Long-end σ : up/down exc.	Expected	Both pass?
5y	1 / 1	4 / 3	1.6	yes ($p > 0.05$)
10y	1 / 0	3 / 2	1.6	yes
20y	1 / 1	3 / 1	1.6	yes

Both sigma choices pass Kupiec at this sample size; the switch to long-end realised vol is motivated by the direct realised-move comparison above, which is the more targeted diagnostic for this book's long-dated exposure.

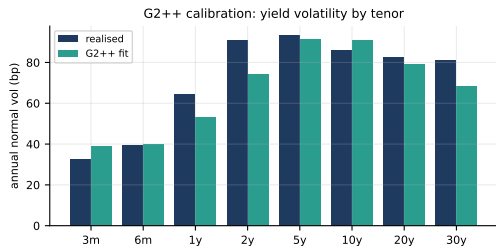
But this book's rate risk sits in one place: 20–30 years



- Par-instrument DV01 (the real curve-building pillars, not an invented grid): **82% of portfolio DV01 sits in the 30y bucket**, vs. 11% at 10y
- BF_0003, the \$500M forward on the 2049 Treasury, dominates – one position, one tenor region
- This is *why* a 2-factor model's main selling point (independent curve twist across many maturities) buys little here: there is effectively one maturity that matters

Same conclusion as a whole-curve PCA would give: concentrated exposure means the fit quality that matters is local (20–30y), not the average fit across the whole curve.

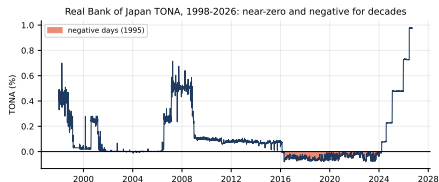
HW1F vs. G2++ on the book: same data, draws, grid and trades



	HW1F	G2++
Fit to realised yield covariance	–	16.1% relative error
Parameters	$a=0.0167$	$\sigma=1.21\%$, $a=0.039$, $\eta=1.44\%$, $b=1.5$ (at bound), $\rho=-0.95$
Portfolio MPE99 (N=2,000, CRN)	baseline \$51.5M	\$50.0M (–3.0%)
By counterparty	–	A –1%, B –3%, C –0%

Benefit of 2-factor: lets the curve twist, not just shift – matters if exposure spans many maturities. **Cost:** a parameter pinned at its bound, 16% fit error, and a 3% move that doesn't justify the complexity here. Same random draws used in both runs (common random numbers), so the –3.0% gap is a real model difference, not sampling noise.

JPY rates: a dedicated Hull-White factor vs. a constant differential



Alternative rejected: JPY names/USDJPY drift off one constant USD-JPY rate differential – cheap, but cannot represent JPY rates moving independently, or negative.

Selected: a second, real, simulated **Hull-White one-factor** model from the JPY OIS curve. TONA has been negative or near-zero for most of 25 years – a floored model cannot represent this. **Mean-reversion calibration**

Route	Fitted a
JPY OIS history	−0.0261
Swaption cube	−0.0374
JGB yield history	−0.0203

– all 3 real routes tried, all invalid:

All negative: JPY vol genuinely *rises* with tenor. Used the lowest admissible value ($a=0.001$, Ho-Lee limit) rather than an invalid fit.

Result: moved portfolio MPE99 attribution \$52.1M → \$51.5M – real,

disclosed, not noise. A CRN on/off test confirms the JPY factor changes no other trade's value beyond that.

Decision register: rate model

Decision	Final choice	Evidence	Trade-off
USD rate model	Hull-White 1F (1-factor LGM)	Book DV01 82% concentrated at 30y; G2++ CRN gap on MPE99 only -3.0% at a 16.1% covariance fit error and a parameter at its bound	No independent curve twist; short-end vol not separately tuned (not this book's risk)
JPY rate model	Hull-White 1F , dedicated factor	All 3 real mean-reversion routes invalid (negative a); $a=0.001$ floor used; attribution moved \$52.1M $\rightarrow$ \$51.5M	One extra simulated factor for a real, if modest, effect
Rate volatility source	Realised 2y–30y Treasury , not overnight SOFR	Matches the actual 10bd long-yield move (15bp vs. 11bp implied); both pass Kupiec	Slightly more model-risk sensitivity to the fitting window

Part IV

Efficient Simulation

Sampling scheme: five methods, exact benchmark numbers

Controlled test (European call, known Black-Scholes truth = 41.077):

Method	RMSE	vs. pseudo-random	ms/trial
Pseudo-random	1.456	1.0x (baseline)	0.30
Antithetic	1.079	1.3x better	0.31
Moment-matched	0.232	6.3x better	0.55
Sobol	0.051	28.8x better	1.65
Latin Hypercube	0.029	50.6x better	0.98

Real 663-dimensional engine check (std. dev. of PFE / median, across repeats): no method wins both – Sobol has the lowest PFE std (2.72M vs. 4.29M pseudo-random), but moment-matched has the lowest median std (0.40M vs. 0.76M pseudo-random); Latin Hypercube is competitive on both (3.73M / 0.48M) without being worst on either. **Selected** Latin Hypercube: best on the controlled test, competitive (never worst) on both real-engine statistics, negligible extra cost.

Latin Hypercube: what it does and does not do

- **Does** stratify every one-dimensional marginal: each risk factor's random draws are forced to cover equal-probability bins, so tails are always visited
- **Does not** stratify the *joint* distribution across all 39 factors and ~ 17 time steps (~ 663 effective dimensions) – interaction noise between factors remains
- **Unbiased**: each draw is still marginally uniform, mapped through the inverse normal CDF, so estimates stay unbiased
- **Cost**: draw generation grows with $N \times \text{dimension}$ – negligible next to repricing here

This is exactly why Latin Hypercube helps a smooth average (EE) more reliably than a 99th-percentile tail (PFE99): the tail is set by interactions between a handful of extreme scenarios, which stratifying single marginals does not directly target.

Scenario count: exact convergence results

Close-out MPE99, relative standard error by bootstrap (peak date 2026-09-20):

Netting set	N=250	N=500	N=1,000	N=2,000	N=3,000
CPTY_A	6.3%	5.1%	3.2%	2.6%	2.5%
CPTY_B	5.1%	4.7%	3.8%	3.5%	3.5%
CPTY_C	5.9%	4.8%	3.8%	3.1%	3.1%
Portfolio	7.1%	4.9%	3.3%	2.2%	2.2%

Portfolio relative SE at $N=2,000$ is 2.2%, extrapolating ($1/\sqrt{N}$) to $\sim 1.4\%$ at $N=5,000$ and $\sim 1.0\%$ at $N=10,000$.

Independent 30,000-scenario pool study (1y-node PFE99, level exposure):

N	1,000	5,000	10,000	20,000	30,000 (pool)
Rel. SE	0.66%	0.29%	0.21%	0.09%	0 (reference)
Est. time (8 cores)	119s	479s	929s	1,829s	2,729s

Selected: $N=5,000$ for standard reporting (good precision, ~ 8 min); $N=10,000$ for sign-off (~ 15 min); beyond $\sim 20,000$, marginal SE improvement per extra 5,000-scenario batch falls below 30% while cost keeps scaling linearly.

Decision register: sampling and path count

Decision	Final choice	Evidence	Trade-off
Sampling scheme	Latin Hypercube	50.6x lower RMSE than pseudo-random on the controlled test; competitive, never worst, on real-engine PFE/median std	No consistent Greeks-noise edge over pseudo-random on PFE99 specifically
Scenario count	$N=5,000$ standard / $10,000$ sign-off	Portfolio rel. SE 2.2% at $N=2,000$, extrapolates to $\sim 1.4\%$ at 5,000; marginal SE gain falls below 30% beyond $\sim 20,000$	$\sim 1.4\%$ residual sampling noise remains in the headline MPE99

Part V

What the Engine Does, and What It Finds

The simulation pipeline

1. **Calibrate:** pull real data → fit every model parameter (Part II)
2. **Draw:** Latin Hypercube random numbers → correlated shocks (Cholesky decomposition of the real 39×39 matrix)
3. **Simulate:** step every risk factor jointly across the full time grid (equities, FX, USD rate, JPY rate)
4. **Reprice:** feed simulated market states into the provided pricers – every trade, every date, every scenario
5. **Aggregate:** net by counterparty → EE / median PFE / PFE99 / MPE, on both exposure conventions
6. **Downstream:** Greeks, stress, CVA/DVA/FVA/KVA, SA-CVA, SA-CCR – all reusing the same simulated paths

Key equations, exactly as implemented

Rates (models/rates.py) – Hull-White one-factor, shifted Ornstein-Uhlenbeck:

$$dr(t) = [\theta(t) - a r(t)] dt + \sigma dW(t), \quad r(t) = x(t) + \alpha(t)$$

Exact-in-distribution step for $x(t)$; $\alpha(t)$ deterministic, fitted to today's curve; closed-form $P(t, T)$.

Equities and FX (models/equity_fx.py) – log-Euler GBM:

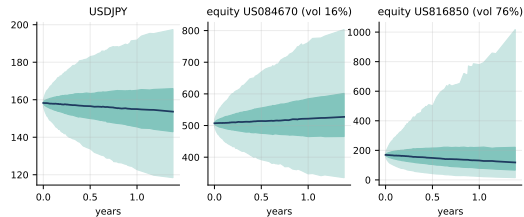
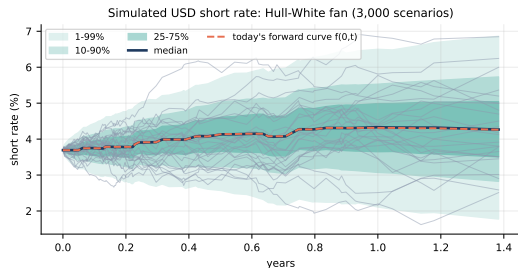
$$d \ln S = (r_{USD} - q - \frac{1}{2} \sigma^2) dt + \sigma dW \quad (\text{USD name}); \quad d \ln X = (r_{JPY} - r_{USD} + \frac{1}{2} \sigma_X^2) dt + \sigma_X dW_X$$

Correlated shocks (simulation/engine.py): $Z = L Z_{indep}$, Cholesky factor of the real 39×39 historical matrix, applied every time step.

Exposure (exposure/spec_exposure.py): $\text{Exposure}_c(t) = \max(V_c(t+10\text{bd}) - V_c(t-1\text{bd}), 0)$, $\text{book} = \sum_c \text{Exposure}_c$.

CVA (exposure/cva.py), Basel MAR50.32 discretisation: $\text{CVA} = \text{LGD} \sum_i \text{DEE}(t_i) \cdot \text{PD}(t_{i-1}, t_i)$.

What the simulated paths actually look like



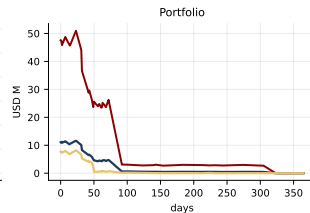
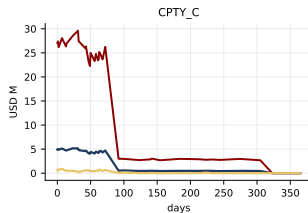
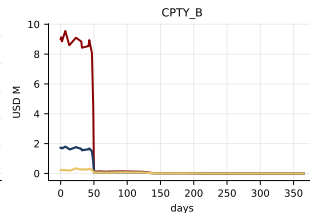
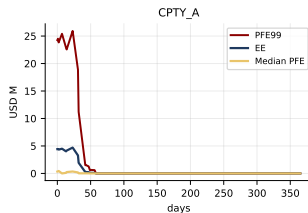
Left: simulated USD short rate (median tracks today's forward curve, since the model is calibrated to reproduce it exactly).

Right: simulated USDJPY and two equities with 1–99% and 25–75% percentile bands.

Headline results (5,000 scenarios, close-out exposure)

	MPE99	Peak EE
CPTY_A	\$25.9M	\$4.7M
CPTY_B	\$9.6M	\$1.8M
CPTY_C	\$29.6M	\$5.2M
Portfolio	\$51.0M	\$11.6M

- CVA \$11.7k close-out / \$247k level
- SA-CVA capital \$0.10M / \$2.23M
- SA-CCR EAD \$260M
- KVA (10% CoC) \$22.4k / \$470k
- Level exposure MPE99: \$201.9M



What drives each counterparty, and book-level Greeks

	Equity $\Delta/+1\%$	FX $\Delta/+1\%$	DV01/ $+1\text{bp}$
CPTY_A	-\$2.25M	\$0	\$31k
CPTY_B	-\$0.96M	+\$0.42M	-\$4k
CPTY_C	-\$1.24M	\$0	\$594k
Book	-\$4.46M	+\$0.42M	\$622k

Rank correlation of exposures: A–B 0.34, A–C 0.33, B–C 0.12

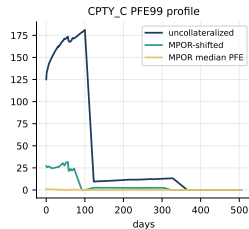
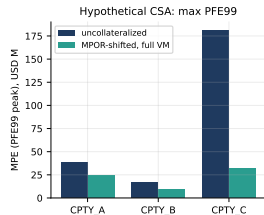
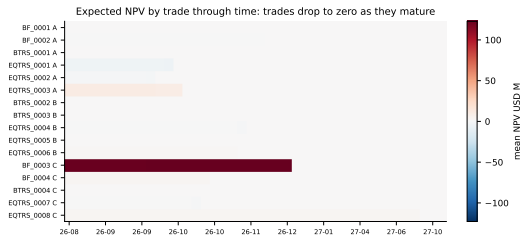
- A is most equity-driven; C is the rate netting set (one \$500M short forward on a 2049 Treasury) – but C also carries meaningful equity delta
- B is the only netting set with material FX delta (holds the 2 JPY-compo swaps)
- Yield-equity correlation ≈ 0.00 in our data – no reliable offset

Why the portfolio is close to additive across counterparties



- Nothing nets *across* counterparties by construction – portfolio MPE99 is a sum of three separate netting sets
- All three counterparties reach their own peak exposure in the same few weeks (before the largest equity baskets mature) – no timing diversification
- Positive rank correlation (0.12–0.34) between netting sets, since every book holds pay-equity swaps
- Result: Portfolio MPE99 (\$51.0M) $\approx$ A+C (\$54.4M), 78% of the simple sum of all three (\$65.1M)

Additional exposure diagnostics



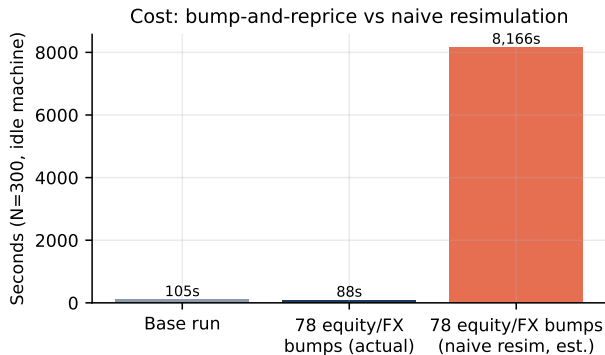
Left: expected NPV by trade – BF_0003 dominates and disappears at its Dec 2026 settlement. Right: peak PFE99 under a hypothetical full-VM CSA, for comparison against the brief's own definition.

Part VI

Sensitivities, Stress, and Validation

Method: common random numbers, and what they buy

- Bumping a risk factor and repricing with **independent** random draws: estimator std. dev. \$245k – too noisy to use
- Reusing the **identical** draws in the base and bumped run (common random numbers, CRN): std. dev. \$1k
- **282x noise reduction** – this, not the sampling scheme, is what actually controls Greeks precision



Equity/FX spot bumps rescale simulated GBM

paths exactly (no resimulation): 78 bumps at N=300 cost 88s actual vs. ~8,166s estimated naive resimulation – **93x cheaper**.

Method comparison: bump-and-reprice vs. pathwise vs. adjoint

Method	Cost	Verdict
Bump, independent draws	baseline, \$245k noise	<i>rejected</i> : noise swamps the signal
Bump, CRN (used for all reported Greeks)	~260s (equity EE, N=300)	<i>Selected</i> : works on black-box pricers, works for quantiles
Pathwise (closed form, equity/FX only)	0.2–0.4s	<i>Selected</i> as fast cross-check: ~1000x faster, agrees to median 0.02% / worst 0.4% vs. bump-and-reprice EE
Adjoint differentiation	would price all Greeks for ~3–5 valuations	<i>rejected</i> : needs pricers rewritten in differentiable form – out of scope, listed as future work

Rate/vol Greeks (13 needed: parallel up/down, 8 buckets, 3 vega) still require a genuine resimulation at ~109s each (N=300, idle machine) – unless the exact analytic method below applies.

Rate DV01: zero-grid bump vs. par-instrument Jacobian – full breakdown

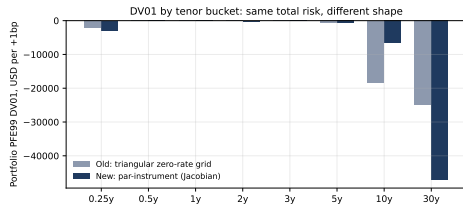
Tenor bucket	Old (zero-grid), USD	New (par-instrument), USD
0.25y	-2,182	-2,892
0.5y	-46	-1
1y	-44	-14
2y	-131	-205
3y	-96	-105
5y	-770	-713
10y	-18,324 (39%)	-6,592 (11%)
30y	-24,988 (54%)	-47,191 (82%)
Sum of buckets	-46,580	-57,713
Independent parallel bump	-46,605	-57,705

Additivity check: both pass. Old sum / parallel = 0.9995; new sum / parallel = 1.0001 – bucket DV01s sum to the independently-computed parallel bump to within 0.05% either way.

Rate DV01: why the new method, and what changed

Old: bump the *fitted* zero curve at 8 hand-picked tenors, triangular interpolation – disconnected from how the curve was actually built.

Selected New: bump the curve's own ~45 native construction pillars (real SOFR-futures-implied points + every Bloomberg-spliced long-end point), grouped into the same 8 report buckets by nearest pillar – the genuine par-instrument



(Jacobian) sensitivity a rates desk would actually compute.

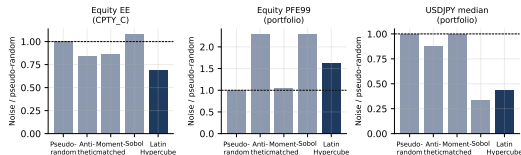
An exact rate Greek, reusing the same simulated draws

In Hull-White 1F, $r(t) = x(t) + \alpha(t)$: $x(t)$ is the simulated random factor (depends only on random draws, a , σ); $\alpha(t)$ is deterministic (depends only on the curve). A curve bump that leaves a , σ fixed – every DV01 bump here – leaves $x(t)$ **identical** to the base run.

- The whole bump effect is one closed-form number per date, added to the base run's own paths
- **No resimulation. No new random draws.**
- Validated against a true independent resimulation: agreement to 10^{-15} relative precision – not an approximation, the same calculation

Separately: does Latin Hypercube itself reduce Greeks noise? Tested directly – mixed, 15–85% less noise on EE deltas, no

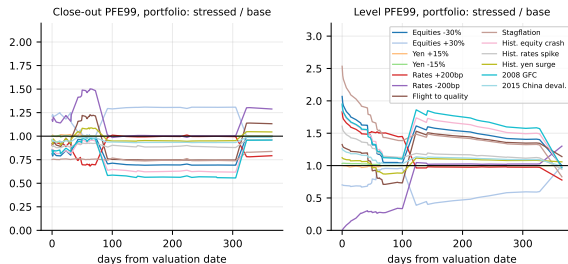
Sampling scheme vs Greeks noise: no consistent winner



consistent benefit on PFE99 deltas:

Stress design: 13 fixed scenarios, same random draws as base

Family	Scenarios	Shock
Hypothetical (8)	EQ_DOWN/UP_30, JPY_STRONG/WEAK_15, RATES_UP/DOWN_200, FLIGHT_TO_QUALITY, STAGFLATION	Round-number shocks to equities, USDJPY, USD curve
Historical (5)	HIST_EQUITY_CRASH, HIST_RATES_SPIKE, HIST_YEN_SURGE, HIST_GFC_2008, HIST_CHINA_DEVAL_2015	Actual moves over a real 10-day window, chosen by worst-window rule

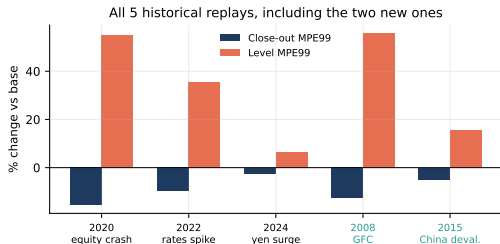


Stress results: how far each scenario moves the headline number

Scenario	Close-out MPE99	Level MPE99
EQ_DOWN_30	−15.3%	+62.3%
RATES_DOWN_200	+22.0%	−69.0%
STAGFLATION	−24.1%	+89.1% (largest level move)
FLIGHT_TO_QUALITY	−2.0%	+19.6%
HIST_EQUITY_CRASH (2020)	−15.4%	+54.8%
HIST_GFC_2008	−12.7%	+55.6%
HIST_CHINA_DEVAL_2015	−4.9%	+15.6%

Close-out exposure is far less sensitive to instantaneous level shocks than level exposure (margin is why). Equity-driven trades react only to equity/FX shocks; bond trades react only to rate shocks, confirming trade structure, not coincidence.

Extending the history: 2008 and 2015, and why it matters

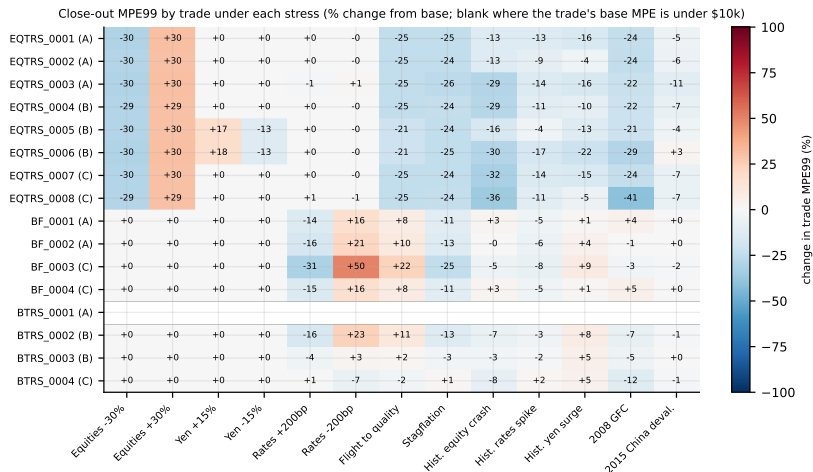


Extended public price history back to 2007 (28 of 37 names covered), found the worst 10-day window inside two named crises:

- **2008 GFC:** Sep 29 – Oct 10, 2008; –23% median / –44% worst single-name return
- **2015 China deval.:** Aug 2015, –6.8% median

Result: both land *inside* the range the existing scenarios already covered – a validation finding, not just more coverage: the pre-existing suite was not blind to the worst episode on record.

Per-trade sensitivity: does every product react as expected



Validation summary: every check has a pass rule

Check	Method / metric	Result
Martingale test	Discounted asset paths should average to today's market price	Pass – no drift or discretisation bug
Parametric VaR benchmark	Delta-normal VaR (bump-and-reprice deltas $\times$ real covariance) vs. full MC P&L, 99%	Parametric \$11.4M vs. MC \$11.7M – ratio 0.98 (expected range 0.3–1.3)
Kupiec backtest, equity (99%, 242 ten-day windows)	Historical exceedances vs. 5,000-scenario prediction	A: 4/2.4, B: 6/2.4, C: 2/2.4 expected – all pass ($p > 0.05$)
Kupiec backtest, rates (99%, 161 windows, 3 tenors)	Same, for the rate factor under both vol sources	0–4 exceedances vs. 1.6 expected at every tenor – all pass
Excel parametric checks	Independent spreadsheet formulas on public data	0.1% (vol calibration), 1–2% (simulated vs. analytic vol), 5–10% (rate vol fit), 7% (equity vega)
DV01 additivity (old method)	Sum of bucket DV01s / independent parallel bump	0.9995 – pass
DV01 additivity (new, par-instrument method)	Same	1.0001 – pass
Exact rate-shift Greek	Analytic path-shift vs. true independent resimulation	Agreement to 10^{-15} relative precision – pass
No cross-counterparty netting	Enforced by construction; portfolio = sum over netting sets	Automated regression test, 168 tests total
2008/2015 stress coverage	New replays vs. range of existing scenarios	Both land inside the pre-existing range – pass

Model risk: every input perturbed, one at a time

Perturbation (base portfolio MPE99 = \$51.5M, N=2,000)	Change
Equity/FX vol $\times 2$	+71.3%
Equity correlations, 30% toward 1	+21.8%
Equity/FX vol $\times 1.25$	+19.3%
Rate–equity dependence, both fall together	+18.2%
Rate volatility $\times 1.25$	+9.4%
Rate–equity dependence, offsetting	–15.3%
Mean reversion $a \times 3$	–4.8%
Two-factor rate model (G2++) instead of one-factor	–3.0%

Conclusion: equity volatility, correlation, and the data-unresolvable rate–equity dependence dominate model risk. The one- vs. two-factor rate-model choice explored at length is, by comparison, one of the *smallest* levers – the strongest argument for keeping the simpler model as the default.

Attribution: every correction, one at a time

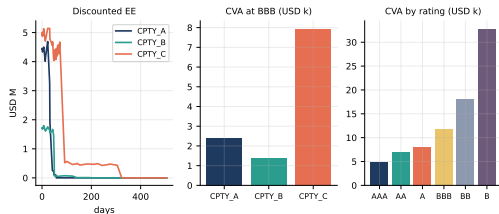
Step	Portfolio MPE99
Earlier version (flat curve beyond 6.5y, SOFR-overnight σ , no JPY factor, USDJPY drift sign error)	\$45.7M
+ Corrected USDJPY drift sign	\$47.8M
+ Long-end volatility fit and 10y-yield correlations	\$52.7M
+ Bloomberg long end spliced onto the USD curve	\$52.1M
+ JPY Hull-White factor (final model)	\$51.5M

No single correction dominates; each is individually disclosed and traceable – the model-risk study above, not any one fix, is the right lens for “how much could the headline number still move.”

Part VII

Credit and Capital

CVA: methodology and results



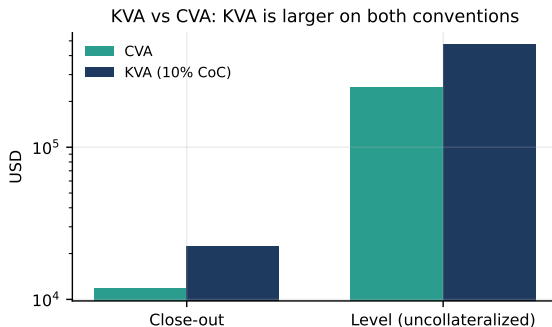
- No CDS available → ICE BofA bond-index OAS proxy by rating (Basel MAR50.32(3)-compliant); all 3 counterparties proxied at BBB
- Credit-triangle hazard-rate survival model:
 $\text{hazard} \approx \text{spread} / (1 - \text{recovery})$, LGD 60%
- CPTY_C close-out CVA \$7.9k; A \$2.4k; B \$1.4k; **total \$11.7k**
- Rating sensitivity: AAA \$5.0k, AA \$7.0k, A \$8.0k, BBB \$11.7k, BB \$18.1k, B \$32.8k

Value delivered: the sensitivity chain (CVA01, the rating table, SA-CVA weighted sensitivities) matters more than the point-in-time number, which is small by construction here.

DVA, FVA, and KVA (extra credit)

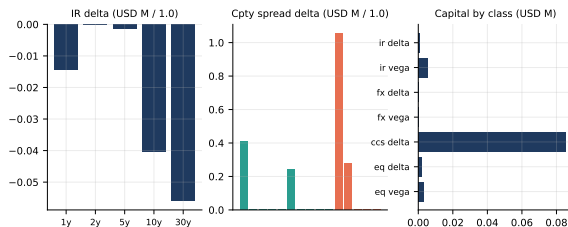
	Close-out	Level
CVA	\$11.7k	\$247k
DVA	\$11.2k	\$7.5k
FVA	\$0.5k	\$240.0k
KVA (10% CoC)	\$22.4k	\$470k

On close-out, positive/negative exposure are nearly symmetric (both 10-day moves from a margined start) so CVA/DVA nearly cancel; on level, the book is net in Capitolis' favour, so CVA/FCA dominate.



KVA: $K(t) = 8\% \times RW \times \alpha \times EPE(t)$, cost-of-capital swept 8/10/12% (none given). Exceeds CVA on both conventions because it scales the whole EPE profile by a fixed ratio, not a small IG default probability – explicitly illustrative.

Regulatory capital: SA-CVA and SA-CCR



- SA-CVA capital: \$0.10M close-out / \$2.23M level (RWA \$1.2M / \$27.8M)
- Dominated by counterparty credit-spread risk on both conventions (5% risk weight on a proxy spread)
- SA-CCR exposure at default: \$260M, dominated by the replacement cost of BF_0003

The kickoff brief flagged *“Risk bonds + CDS data for new sample trades”*. Delivered:

- A sample risky-bond forward (25M notional, 5% BBB corporate; not part of the ESF book)
- Priced and simulated with a real issuer credit curve, re-anchored at every simulated node
- Same hazard-rate survival model and bond-index proxy construction as the counterparty CVA

Limitation, disclosed: no real CDS quotes were obtainable for a hypothetical issuer, so the issuer curve is a bond-index proxy, not actual CDS – delivered in spirit, not the full ask.

Part VIII

Decisions, Limitations, and Conclusions

Full decision register

Decision	Final choice	Evidence	Trade-off
USD rate model	Hull-White 1F (1-factor LGM)	G2++ CRN gap only -3.0% at 16.1% fit error, parameter at bound; book DV01 82% at 30y	No independent curve twist
JPY rate model	Hull-White 1F, dedicated factor	All 3 real α -routes invalid; attribution \$52.1M → \$51.5M	One extra simulated factor
Rate volatility source	Realised 2y–30y Treasury	Matches actual 10bd long-yield move (15bp vs. 11bp implied); both pass Kupiec	More model-risk sensitivity to window choice
Correlation	Full 39 × 39 Cholesky	PCA is slower (2.08s vs. 1.91s) with up to 1.7% extra error	Estimation noise across 741 pairs
Sampling scheme	Latin Hypercube	50.6x lower RMSE vs. pseudo-random on controlled test	No consistent Greeks-noise edge on PFE99
Path count	$N=5,000$ / 10,000 sign-off	Rel. SE 2.2% at 2,000, extrapolates to ~1.4% at 5,000	~1.4% residual sampling noise
Exposure convention	Close-out primary, level secondary	Brief's own definition; level is a 4x larger no-margin bound	Real CSA terms (threshold/MTA/IM) unknown
Greeks method	Bump-reprice + CRN; path-wise cross-check	282x noise reduction; pathwise 1000x faster, 0.02–0.4% agreement	Rate/vol Greeks still need resimulation (unless exact method applies)
Rate DV01 attribution	Par-instrument Jacobian buckets	82% vs. 54% of risk correctly placed in 30y bucket	More construction-pillar bookkeeping
Credit spread source	ICE BofA rating-index OAS proxy	Basel MAR50.32(3)-sanctioned fallback; no CDS available	No sector/region discrimination, no wrong-way risk
KVA	Parametric, CoC grid	Completes xVA extra-credit ask	Not a real capital methodology

Why this engine

Accurate

- Parametric VaR vs. MC: ratio 0.98
- Kupiec backtest: all pass (equity and rates, 95% and 99%)
- DV01 additivity: 0.9995 / 1.0001

Fast

- Equity/FX bumps: 93x cheaper than naive resimulation
- Pathwise delta: $\sim 1000\times$ faster than bump-and-reprice
- Exact rate-shift Greek: no resimulation at all

Explainable

- Every parameter traced to a named, dated data source
- Every model choice backed by a quantified trade-off (decision register above)

Stable

- CRN cuts Greeks noise 282x (\$245k $\rightarrow$ \$1k)
- Exact rate-shift Greek agrees with resimulation to 10^{-15}
- G2++ challenger within 3% of production on MPE99

Reproducible

- Same seed $\rightarrow$ identical paths and NPVs
- 168 automated regression tests
- Every number in this deck traces to a committed result file

Accurate + fast + explainable + stable + reproducible – each claim backed by a number in this deck.

Limitations and model risk

Area	Limitation	Effect	Mitigation / next step
Volatility	Realised (3y), constant; no smile or clustering	Exposure could be understated in a vol spike (model-risk: vol $\times 2$ moves MPE99 +71%)	Implied vols; stress overlay already tests a regime shift
Credit	BBB proxy for all counterparties; flat term shape; independence of exposure and default	CVA scales with the assumed spread; no wrong-way risk	Real ratings/CDS; sector-specific proxy; WWR test
Rates	One factor in production	Curve-twist (slope) risk not captured	G2++ challenger kept and re-tested; re-visit if slope exposure grows
Capital	KVA uses an assumed cost-of-capital grid, no real methodology	KVA number is illustrative only	Real capital methodology and CoC from Capitolis
Validation Monte Carlo	One valuation date; no rolling exposure backtest MPE99 (a tail quantile) converges slowly	Model error over time unmeasured $\sim 1.4\text{--}2.2\%$ relative spread at $N=5,000$	Rolling backtest across multiple dates Sign-off runs at $N=10,000$

Decision made: production configuration

Production configuration

- **Rates:** Hull-White 1F, $a=0.0167$, $\sigma=0.96\%$; JPY Hull-White 1F, $a=0.001$; G2++ kept as challenger
- **Equities / FX:** correlated GBM, full 39-factor Cholesky, quanto drift for JPY names
- **Simulation:** Latin Hypercube, $N=5,000$ (sign-off 10,000), common random numbers for every Greek
- **Exposure:** close-out 10bd with $t-1$ bd VM, per netting set, book = sum: **MPE99 \$51.0M, peak EE \$11.6M, CVA \$11.7k**

Monitor

- Book slope exposure (would trigger a G2++ re-test)
- Volatility regime (implied vs. realised)
- Credit proxy vs. any real counterparty CDS

Next

- Real ratings/CDS, own credit and funding spread
- Real capital methodology and CoC for KVA
- Sector-specific credit-spread proxy; rolling exposure backtest

Feedback tracker: both rounds, all 11 items

Earlier meeting

1. Robust stress testing [Selected](#)
2. Sensitivities: method, cost, faster alt. [Selected](#)
3. PCA explainability and speed [Selected](#)
4. Final results explanation [Selected](#)
5. CVA walkthrough and set-up [Selected](#)

This meeting

6. 2008 and more testing periods [Selected](#)
7. LHS to calculate sensitivities [Selected](#)
8. KVA addition [Selected](#)
9. CVA is for sensitivities [Selected](#)
10. DV01 via a Jacobian [Selected](#)
11. Excel parametric vol-bump test [Selected](#)

Detail and results for every item are in the sections above.

Thank you

Questions and discussion